A -> A_PERF FOR A = F_P[T] IS FLAT AND FORMALLY UNRAMIFIED BUT NOT FORMALLY ETALE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Author source typo kept verbatim under content zero-change: '$L_{A/\mathbb{F}_P}$' (capital P) occurs once in the cotangent triangle sentence; also one sentence ends with the citation and no period. No rewrite.

Editorial note. The label lemma-perfect-closure-polynomial-ring names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Perfect closure as a colimit of Frobenius towers; faithful flatness of Frobenius on F_p[T] via the free basis {1, T, ..., T^(p-1)}
- Formal unramifiedness via vanishing of Omega (More on Morphisms, formally-unramified-differentials)
- The cotangent complex distinguished triangle (Cotangent, section-triangle) and the naive cotangent criterion for formal smoothness (More on Morphisms, NL-formally-smooth; Cotangent, relation-with-naive-cotangent-complex)

Source reference. Original lemma 0G60: sources/stacks-chow/tags.txt, source lines 20180–20180. Prerequisite pointer invoked at 3458; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08RB: sources/stacks-chow/tags.txt, source lines 11089–11089. Prerequisite pointer invoked at 3473; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08R5: sources/stacks-chow/tags.txt, source lines 11083–11083. Prerequisite pointer invoked at 3456; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 08QR: sources/stacks-chow/tags.txt, source lines 11069–11069. Prerequisite pointer invoked at 3466; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D0L: sources/stacks-chow/tags.txt, source lines 16316–16316. Prerequisite pointer invoked at 3471; it names a standard fact of a

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sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02H9: sources/stacks-chow/tags.txt, source lines 3256–3256. Prerequisite pointer invoked at 3448; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $A = \mathbb{F}_p[T]$ be the polynomial ring in one variable over $\mathbb{F}_p$. Let A_{perf} denote the perfect closure of A. Then $A \to A_{perf}$ is flat and formally unramified, but not formally étale.*

Proof. Note that under the Frobenius map $F_A : A \to A$, the target copy of A is a free-module over the domain with basis $\{1, T, \ldots, T^{p-1}\}$. Thus, F_A is faithfully flat, and consequently, so is $A \to A_{perf}$ since it is a colimit of faithfully flat maps. Since A_{perf} is a perfect ring, the relative Frobenius $F_{A_{perf}/A}$ is a surjection. In other words, $A_{perf} = A[A^p_{perf}]$, which readily implies $\Omega_{A_{perf}/A} = 0$. Then $A \to A_{perf}$ is formally unramified by More on Morphisms, Lemma 02H9

It suffices to show that $A \to A_{perf}$ is not formally smooth. Note that since A is a smooth $\mathbb{F}_p$-algebra, the cotangent complex $L_{A/\mathbb{F}_P} \simeq \Omega_{A/\mathbb{F}_p}[0]$ is concentrated in degree 0, see Cotangent, Lemma 08R5. Moreover, $L_{A_{perf}/\mathbb{F}_p} = 0$ in $D(A_{perf})$ by Cotangent, Lemma 0G60. Consider the distinguished triangle of cotangent complexes

$$L_{A/\mathbb{F}_p} \otimes_A A_{perf} \to L_{A_{perf}/\mathbb{F}_p} \to L_{A_{perf}/A} \to (L_{A/\mathbb{F}_p} \otimes_A A_{perf})[1]$$

in $D(A_{perf})$, see Cotangent, Section 08QR. We find $L_{A_{perf}/A} = \Omega_{A/\mathbb{F}_p} \otimes_A A_{perf}[1]$, that is, $L_{A_{perf}/A}$ is equal to a free rank 1 A_{perf} module placed in degree -1. Thus $A \to A_{perf}$ is not formally smooth by More on Morphisms, Lemma 0D0L and Cotangent, Lemma 08RB. □